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④ Epsilon (ϵ) - NFA 5.3,

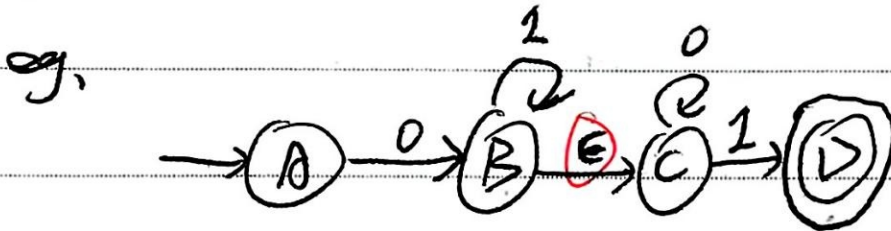
ϵ -NFA

↳ empty symbols

$\{Q, \epsilon, q_0, \delta, F\}$ same as ~~NFA~~ ^{NFA}

$$\delta : Q \times \Sigma \rightarrow 2^Q$$

for ϵ -NFA: $\delta : Q \times \Sigma \cup \epsilon \rightarrow 2^Q$

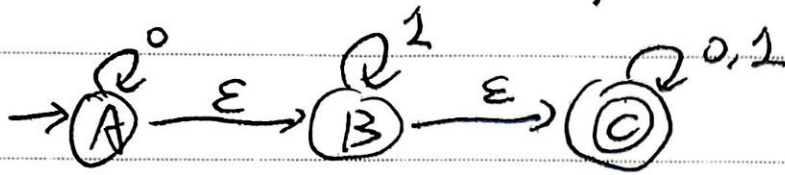


Every State on ϵ goes to it self



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Conversion of E-NFA to NFA

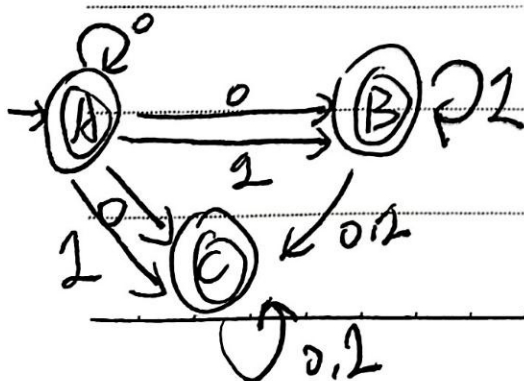
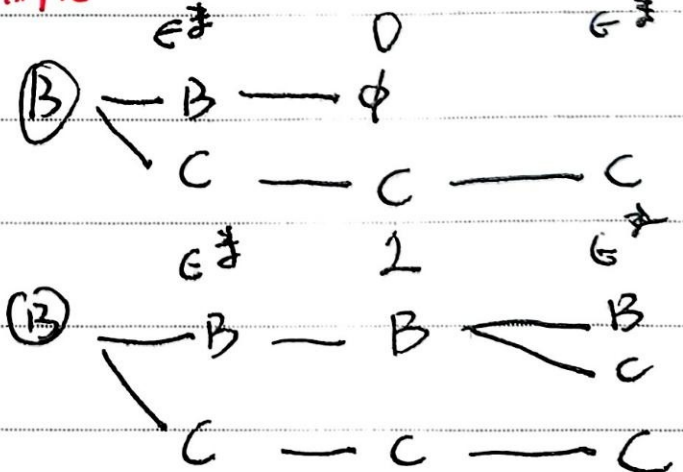
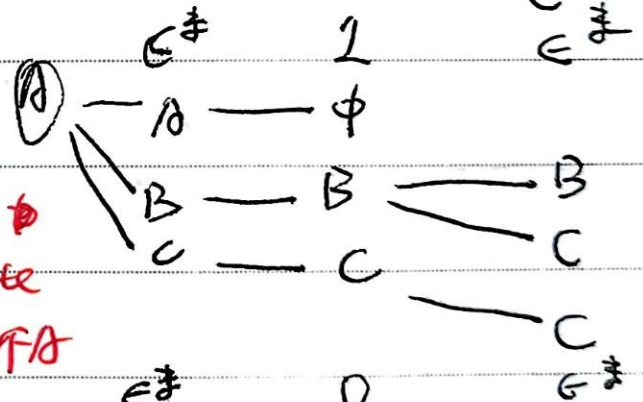
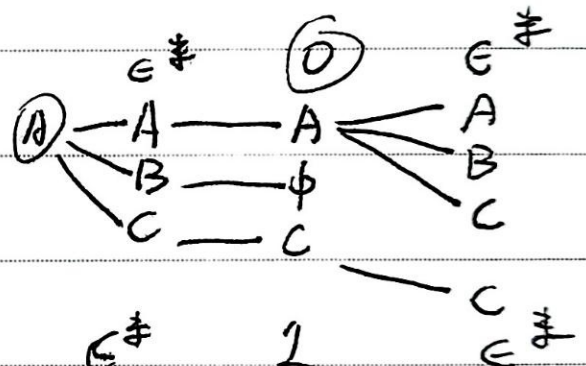


Starts ϵ^* input ϵ^*

E. Closure (ϵ^*) - All the states that can be reached from a particular state only by seeing the ϵ symbol

	0	1
$\rightarrow A$	{A, B, C}	{B, C}
B	{C}	{B, C}
C	{C}	{C}

all can be seen with ϵ^* to final state in E-NFA





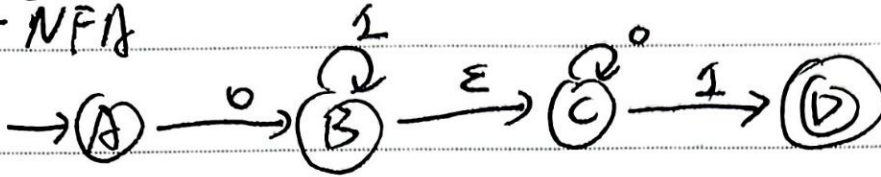
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43/36 Conversion of E-NFA to MFA Example 1

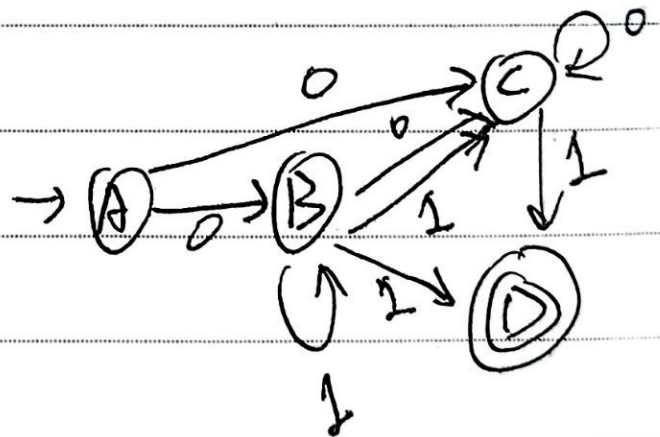
E-NFA



	0	1
→A	{B, C}	∅
B	C	{B, C, D}
C	C	∅
D	∅	∅

	ε	0	1
A	A → B	B → C	∅
B	∅	C → C	∅
C	∅	C → C	∅
D	∅	∅	∅

	ε	1	ε
A	A → ∅	∅	∅
B	B → B	B → C	∅
C	C → ∅	∅	∅
D	∅	∅	∅





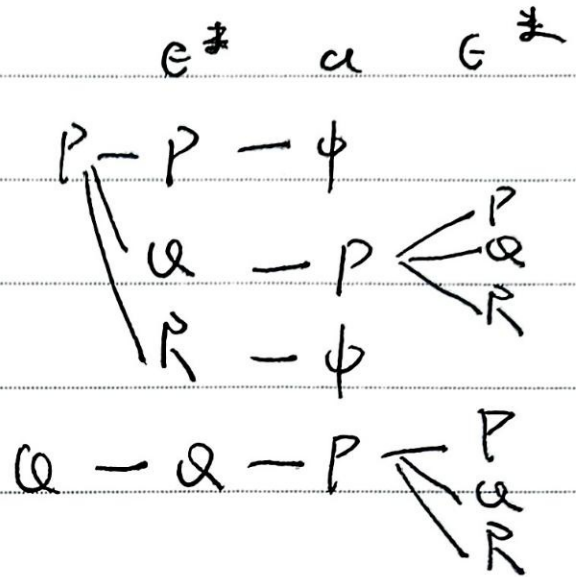
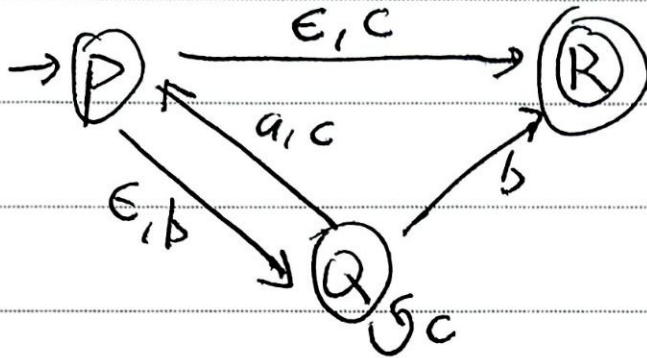
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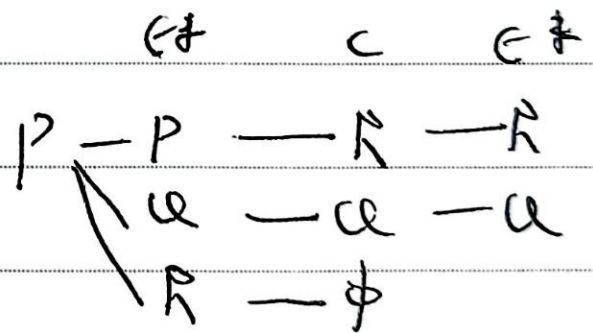
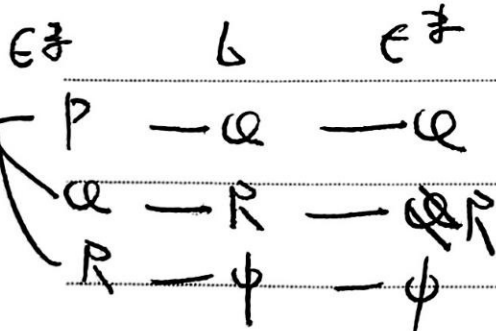
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Example 2



	a	b	c
P	{P, Q, R}	{Q, R}	{R, Q}
Q	{P, Q, R}	{R}	{Q}
R	ϕ	ϕ	ϕ

R - R - ϕ



Q - Q - Q - Q

R - R - ϕ

R - R - ϕ

